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EMTL 24EC2207 CO-3 Home Assignment 2400040454

1) In free space, the magnetic field of an EM wave is given by $H = 0.4\omega\varepsilon_0 \cos(\omega t - 50x) a_z$ A/m. then Compute the electric field, E and displacement Current density.

Sol: Given Data:

$H = 0.4\omega\varepsilon_0 \cos(\omega t - 50x) a_z$ A/m

For a uniform plane wave : $E = \eta_0 H$ $[\because \eta_0 = 377\Omega]$

Direction rule for propagation $(\vec{E} \times \vec{H})$ along x

$$E = \eta_0 H$$

$$= 377\Omega\,(0.4\omega\varepsilon_0 \cos(\omega t - 50x))(-a_y)$$

$$= -150.8\,\omega\varepsilon_0 \cos(\omega t - 50x) a_y$$

$$\boxed{E = -150.8\,\omega\varepsilon_0 \cos(\omega t - 50x) a_y \text{ V/m}}$$

Displacement Current Density $J_d = \frac{\partial D}{\partial t}$

$[D = \varepsilon_0 E]$

$$J_d = \frac{\partial \varepsilon_0 E}{dt}$$

$$J_d = \varepsilon_0 \frac{\partial}{\partial t} E$$

$$J_d = \varepsilon_0 \frac{\partial}{\partial t}\left(-150.8\,\omega\varepsilon_0 \cos(\omega t - 50x) a_y\right)$$

$$= \varepsilon_0(-150.8\,\omega\varepsilon_0)(-\omega \sin(\omega t - 50x)) a_y$$

$$= 150.8\,\omega^2\varepsilon_0^2 \sin(\omega t - 50x) a_y$$

$$\boxed{J_d = 150.8\,\omega^2\varepsilon_0^2 \sin(\omega t - 50x) a_y \text{ A/m}^2}$$

2) An electric field in a medium which is Source free given by $E = 1.5\cos(10^8 t - \beta z) a_x$ V/m, where Em is the amplitude of E, ω is the angular frequency and β is the phase Constant. then Solve B, H, D Assume that relative permitivity and permability are 1.

sol:

Given Data

$E = 1.5\cos(10^8 t - \beta z)\, a_x$ V/m

amplitude $E_m = 1.5$ V/m

phase Constant $= \beta$

Relative permitivity $\epsilon_r = 1$ } → free space.

Relative permeability $\mu_r = 1$

Intrensic impendence $\eta = 377\,\Omega$

1) Magnetic field H :

$$H = E/\eta$$

$$H = \frac{1.5\cos(10^8 t - \beta z)}{377}$$

$$H = 3.98\times10^{-3}\cos(10^8 t - \beta z)$$

$$\boxed{H = 3.98\times10^{-3}\cos(10^8 t - \beta z)\, a_y \text{ A/m}}$$

2) Magnetic Flux Density :→

$$B = \mu_0 H \qquad [\because \mu_0 = 4\pi\times10^{-7}]$$

$$B = (4\pi\times10^{-7})\{3.98\times10^{-3}\cos(10^8 t - \beta z)$$

$$= 5.0\times10^{-9}\cos(10^8 t - \beta z)$$

$$\boxed{B = 5.0\times10^{-9}\cos(10^8 t - \beta z)\, a_y \text{ T}}$$

3) Electric flux Density :→

$$D = \epsilon_0 E$$

$$[\because \epsilon = 8.854\times10^{-12}]$$

$$= 8.854\times10^{-12}(1.5)\cos(10^8 t - \beta z)$$

$$\boxed{D = 1.33\times10^{-11}\cos(10^8 t - \beta z)}$$

When the amplitude of the magnetic field in a plane wave is 2A/m, then Compute the magnitude of the electric field when the plane wave in free space also determine the magnitude of electric field when the wave propagation in a medium which is characterized by $\sigma = 0$, $\mu = \mu_0$, $\epsilon = 4\epsilon_0$!

Given Data :→

$H = 2A/m$

plane wave in free space

$E = \eta H$ $\quad \therefore [\eta = 377\Omega]$

$E = 377 \times 2$

$$\boxed{E = 754 V/m}$$

Wave in medium

$\sigma = 0,\ \mu = \mu_0,\ \varepsilon = 4\varepsilon_0$

we know that

$$\eta = \sqrt{\frac{\mu}{\varepsilon_0}} = \sqrt{\frac{\mu_0}{4\varepsilon_0}} = \frac{1}{2}\sqrt{\frac{\mu_0}{\varepsilon_0}} \qquad \therefore \left[\sqrt{\frac{\mu_0}{\varepsilon_0}} = \eta = 377\right]$$

$$\eta = \frac{377}{2} = 188.5\Omega$$

$E = \eta \times H$

$E = 188.5 \times 2$

$$\boxed{E = 377 V/m}$$

4) An elliptically polarized wave has an electric field $E = \sin(\omega t - \beta z)a_z + 2\sin(\omega t - \beta z + 75^\circ) a_y$ V/m. Then Compute the power per unit area Conveyed by the wave in free space.

Given Data :

$$E = \sin(\omega t - \beta z)a_x + 2\sin(\omega t - \beta z + 75^\circ) a_y \text{ V/m}.$$

We know that power per unit area

$$P_{avg} = \frac{E_{rms}^2}{\eta} = \frac{E_{ox}^2 + E_{oy}^2}{2\eta}$$

$$= \frac{1^2 + 2^2}{2(377)}$$

$\therefore E_{ox} \rightarrow$ amplitude of x Component

$E_{oy} \rightarrow$ amplitude of y component

$$\boxed{P_{avg} = 6.63 \times 10^{-3} W/m^2}$$

5) If the electric field strength E of a uniform plane wave in free space is given by $E = 2\cos\omega(t - z/v_0)\,a_y$ v/m, then Compute the magnetic field intensity !

Sol: Given that

$$E = 2\cos\omega(t - z/v_0)\,a_y \text{ v/m}$$

Relationship between E & H

$$E = \eta \times H$$

$$H = E/\eta$$

$$H = \frac{2}{377}\cos\omega\left(t - \frac{z}{V_0}\right)$$

$$H = 5.3\times10^{-3}\cos\omega\left(t - \frac{z}{V_0}\right)$$

$$\boxed{H = -5.3\times10^{-3}\cos\omega\left(t - \frac{z}{V_0}\right) a_x \text{ A/m}} \quad \therefore [H = -a_x]$$

6) Compute electric flux density and Volume charge density if the electric field $E = x^2 a_x + 2y^2 a_y + z^2 a_z$ v/m in a medium whose $\varepsilon_r = 2$!

Given Data :

$$E = x^2 a_x + 2y^2 a_y + z^2 a_z \text{ v/m} \quad \varepsilon_r = 2$$

electric flux Density $D = \varepsilon E$

$$D = \varepsilon_r \varepsilon_0 E$$

$$= 2\times 8.85\times10^{-12}(x^2 a_x + 2y^2 a_y + z^2 a_z)$$

$$\boxed{D = 1.7708\times10^{-11}(x^2 a_x + 2y^2 a_y + z^2 a_z)\,\text{c/m}^2}$$

Volume charge Density ρ_v

$$\rho_v = \nabla \cdot D$$

$$\rho_v = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\boxed{\rho_v = 3.54\times10^{-11}x + 7.08\times10^{-11}y + 3.54\times10^{-11}z \ \text{c/m}^3}$$